COMP7095 Project: Case Study of Big Data Management

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February 23, 2026

Important Milestones:

- Group Registration Deadline: **17:00 Mar. 10, 2026**
- Project proposal Deadline: **17:00 Mar. 24, 2026**
- Project Report Deadline: **17:00 April 14, 2026**

1 Introduction

In this project, you are required to complete a project about big data management. You will jointly work with your teammates to complete this project that contains the components: (1) the motivation, (2) details on the design/implementation of your ideas, (3) proposed algorithms/schemes/solutions in detail (e.g., including pseudocode, workflow, flow chart, or neural network structures), and (4) simulations/experiments to validate the proposed ideas.

Upon completion of this project, you need to submit a comprehensive technical report written in English. Moreover, you are also required to present your project in a short video (within 5 minutes).

2 Objectives

After completing this project, you shall obtain some basic capabilities to conduct research and development, especially for *identifying* problems, *solving* the problems with proposed solutions, *implementing* your solutions, *writing* a report, and *presenting* your project.

3 Project Criteria

Your project will be graded primarily based on the following weighting scheme:

- Project grouping and project proposal (10%)
- Project report (60%)
- Evaluation of reproducibility artifacts on executable program codes with data (20%)
- Presentation and demonstration (10%)

4 Specification

4.1 Topics

You can choose from one of the following categories to conduct comprehensive R & D work (but not limited to them):

1. Retail and shopping data;
2. Transportation data;
3. E-commerce data;
4. Financial data;
5. Social-network data;
6. Movie data;
7. Business data;
8. Stock data;
9. Environment data;
10. and other application data.

4.2 Tasks

As indicated in Lecture 1, big data management typically consists of the following four stages: (1) Data collection (about how to obtain the data); (2) Data cleaning or preprocessing (if necessary); (3) Data processing and storage (you may consider HDFS, PySpark, and SQL/NoSQL approaches); and (4) Data visualization and analytics (you may consider using data visualization tools/analytics tools learned from our classes to further process your data to obtain some intuitive insights).

Your data management system may consist of all the above four stages, depending on your data. You should adopt at least THREE approaches/tools that you have learned from our classes in your system. You need to explicitly explain which of the four stages are included in your project and justify the reason.

4.3 Data sources

You may obtain data sources via Kaggle (`https://www.kaggle.com/`), Google Dataset Search (`https://datasetsearch.research.google.com/`), Datahub.io (`https://datahub.io/collections`), Hong Kong Open Data (`https://data.gov.hk/en/`), Tianchi Dataset (`https://tianchi.aliyun.com/dataset`), and other data sources (even by crawling the data by yourself). You need to provide details on the data sources, such as URLs of data sources as well as the method to obtain the data.

4.4 Project Grouping

Projects should be done in groups of **THREE to FIVE students**. Please specify your work distribution in the final project report. The deadline for grouping is **17:00 Mar. 10, 2026**. For the students who cannot form a group, TAs will help to assign a group manually after the grouping deadline.

4.5 Project Proposal

After selecting a topic, you are required to write a brief proposal (at least TWO A4 double-column pages and at most FOUR A4 double-column pages in ACM format). The proposal should include the title, a brief abstract of the proposed project, an introduction to your R& D proposal with some technical details, and necessary references. Each group should turn in the proposal softcopy in PDF format in the course Moodle by the due date specified above. In the proposal, you should provide information including Group IDs, student names, student IDs, and email addresses. In particular, your proposal should discuss the following issues:

- How does the project relate to the course material?
- Why do you choose this topic?
- What questions do you want to address?
- Which methods/solutions will you use to address the above questions?
- What are you planning to do next?
- An overview of what work must be done and how it will be divided amongst the group (i.e., specifying the role of each group member).

You do not necessarily answer the above questions in a Q/A form, but you can implicitly address them in your written proposal.

The deadline for submitting the project proposal is **17:00 Mar. 24, 2026**.

5 Project Final Report

5.1 Content

Your final report shall include:

1. Introduction and contributions made by your group mates;
2. Related tools to complete your project;
3. The pipeline flow of your data management system and the overall architecture of your system;
4. Design and implementation details (including at least pseudocode, the working flow of algorithms, and flow charts);
5. Numerical results based on simulations or realistic experiments (data visualization is necessary);
6. Conclusion;
7. References; and
8. Appendices (if any)[1].

[1]Feel free to include additional material in your report if you wish to include, e.g., additional experimental results and a comprehensive literature survey. Your supplementary material should be appended after the 10 pages (the normal content should be presented within 10 pages).

Kindly note that you should not simply copy/paste your program codes or screenshots of your program codes in the final report. Your program codes should be submitted separately as Reproducibility Artifacts (see Section 7).

Requirements of your report:

- Your report should be fully written in English.
- You should write the report in your own words - DO NOT directly copy words/texts from other papers or technical blogs.
- You should properly cite references or other related studies if necessary.
- You should explicitly explain figures or tables when you refer to them.

Any violation of the above rules may lead to a penalty for your score.

5.2 Format

You should write your report as if you are writing a conference paper. You are required to use the ACM Proceedings Template at this link: `https://www.acm.org/publications/proceedings-template`

You could use either latex or MS Word to write the final report. However, your group needs to submit a PDF file in the course Moodle before the final report deadline. The page requirement for the normal content is at most 10 double-column pages (including references), while it should have at least 8 double-column pages. Each page should be formatted in single-space and double-column mode with 10-point font size. If you have not written such a report before, you may want to take a look at research papers from corresponding ACM conferences.

5.3 Submission

Please submit the completed project report, the datasets, and source codes in the course Moodle according to the given deadline. The datasets and the methods (codes) that you have used are also suggested to be submitted though they should be in limited size (e.g., 50 MB) due to the limitation of the course Moodle. You also need to specify the compile and execution methods for your codes (i.e., a README file). If your dataset/codes are larger than 50 MB, you may offer a URL link to let us download your dataset/codes. Details about the source codes and the data can be referred to Section 7.

The submission deadline is **17:00 April 14, 2026**.

6 Project Presentation Video

Each group should submit a video no longer than 5 minutes to present your project. Your video should present the motivation, the main idea of your proposed solution, and the main results.

The video file MUST meet the following requirements:

- Landscape orientation 16:9
- Resolution: 1280 x 720 (=16:9 ratio, 720p)
- FPS: 30 frames/sec
- In .MP4 format optimized for streaming

- The Video Codec MUST be H.264
- The Audio Codec MUST be AAC
- Max file size: 500MB

The deadline for submitting the project presentation video is **17:00 April 14, 2026**.

7 Evaluation of Reproducibility Artifacts

Artifacts comprise software, datasets, environment configuration, mechanized proofs, benchmarks, test suites with scripts, etc. A complete artifact package must contain (1) the computational artifacts and (2) instructions/documentation describing the contents and how to use them. Regarding the artifact, in particular the code and the datasets, scripts should be provided to support the compilation, deployment, and execution to support the reproducibility of experiments.

The artifact description must be included in a README file along with the artifact, and it must include the following aspects:

- Artifact Identification: (i) the report's title, (ii) the group No, the student names, and student IDs, and (iii) an abstract describing the main contributions of the project and how the role of the artifact in these contributions. The abstract may include a software architecture or data models and its description to help to understand the artifact and a clear description of to what extent the artifact contributes to the reproducibility of the experiments in the report.
- Artifact Dependencies and Requirements: (i) a description of the hardware resources required, (ii) a description of the operating systems required, (iii) the software libraries needed, (iv) the input dataset needed to execute the code or when the input data is generated, and (v) optionally, any other dependencies or requirements. Best practices to facilitate the understanding of the descriptions indicate that unnecessary dependencies and requirements should be suppressed from the artifact.
- Artifact Installation and Deployment Process: (i) the process description to install and compile the libraries and the code, and (ii) the process description to deploy the code in the resources. The description of these processes should include an estimation of the installation, compilation, and deployment times.
- Reproducibility of Experiments: (i) a complete description of the experiment workflow that the code can execute, (ii) an estimation of the execution time to execute the experiment workflow, (iii) a complete description of the expected results and an evaluation of them, and most importantly (iv) how the expected results from the experiment workflow relate to the results found in the report. Best practices indicate that the expected results from the artifact should be in the same format as the ones in the submitted report to facilitate the understanding of the scope of the reproducibility. For instance, when the results in the report are depicted in a graph figure, ideally, the execution of the code should provide a (similar) figure (there are open-source tools that can be used for that purpose, such as gnuplot and Matplotlib).